

BERA v1.1 — D(t) Canonical Definition + Gate Formalization (A · B · C · D)

Instrument ID: ERES-BERA-2026-001 **Version:** v1.1 — DRAFT addendum to v1 **License:** CCAL v2.1 **Author:** Joseph Allen Sprute, ERES Institute for New Age Cybernetics **ORCID:** 0000-0001-9946-3221 **Status:** Proposed for JAS canonical lock + MIEVM audit. Cross-domain claim (§1.4) gated on Pellis sign-off.

Resolves v1 open items #2 (D(t) functional form) and adds the Gate family. Locking authority is JAS; this document supplies the candidate forms for that lock.

PART 1 — D(t): the disruption index (proposed canonical lock)

D(t) is a **rating-tier** quantity (continuous, tuning). Per the gate/rating split (BERA v1 §2), D(t) does **not** authorize anything by itself; an onset crossing may inform tuning, but admit/refuse stays with the gate family in Part 2.

1.1 Per-channel coherence

For each observable channel $k \in \{\text{HRV}, \text{EDA}, \text{RESP}, \text{VOC}\}$, define a coherence sub-score over the window ending at t:

$$c_k(t) \in [0, 1] \quad \text{where } 1 = \text{maximally coherent}, 0 = \text{fully decohered}$$

Each channel's metric is its own sub-spec (e.g. HRV coherence ratio, respiratory regularity, EDA tonic stability, VOC stability). The only lock-level requirement is the range and orientation: $\overline{c_k} \in [0, 1]$, 1 = coherent.

1.2 Aggregate coherence

$$C(t) = \sum_k w_k \cdot c_k(t), \quad \text{with integer weights } w_k, \quad \sum_k w_k = W_{\text{sum}}$$

Weights are integers and the sum uses integer cross-multiplication (same determinism rule as RATE; no floating-point divergence across platforms). C(t) is then normalized to [0,1] by W_{sum} .

1.3 The disruption index (LOCK CANDIDATE)

Let $\overline{C_0}$ = baseline aggregate coherence (mean C over the reference / resting block — block A in the ERES-PELLIS A-B-A-B-A-B design).

$$D(t) = \begin{cases} 0 & \text{if } C(t) \geq C_0 \\ \text{clamp}_{[0,1]}((C_0 - C(t)) / C_0) & \text{otherwise} \end{cases}$$

- **D(t) = 0** — at or above baseline coherence (resonant; no disruption)
- **0 < D(t) < 1** — fractional coherence loss from baseline
- **D(t) → 1** — coherence collapse

D(t) is bounded [0,1], deterministic, and replayable, consistent with the rest of BERA.

1.4 Disruption onset (the object of the Pellis hypothesis)

Onset time **t*** is the first t where a smoothed rate condition holds:

$$dD/dt \mid_{\text{smoothed}} \geq \theta_D \quad \text{AND} \quad D(t) \geq D_{\min}$$

θ_D (rate threshold) and $D_{\min}$ (floor) are tunable parameters, not constants.

HYPOTHESIZED — conceptual, not engineered (BERA-PSO correspondence): The working hypothesis is that the **functional form of D(t) as $t \rightarrow t^*$** in a physiologically stressed system shares its shape with spectral-collapse onset in a confined fusion plasma. The proposed *named* mechanism for a shared form is **critical-transition early-warning theory** (critical slowing down): rising lag-1 autocorrelation and rising variance of the coherence residual $(C(t) - C_0)$ preceding t^* . This is a hypothesis to be tested under the ERES-PELLIS protocol, not a demonstrated result, and it is tagged as such wherever it appears. Any deposit touching this correspondence is gated on Pellis sign-off.

1.5 What is being locked vs. left open

- **LOCK (this document):** the form of D(t) in §1.3; D(t) as rating-tier (non-gating); the onset condition shape in §1.4.
- **OPEN (v2):** per-channel metric sub-specs (§1.1); integer weight vector w_k ; θ_D and $D_{\min}$ defaults; the early-warning estimator window for autocorrelation/variance.

PART 2 — The Gate family (A · B · C · D)

Each gate is a **gate-tier** invariant: it admits or refuses, and refusal raises `LockViolation(<gate>)`.

In the BIND/REFUSE/VEILED idiom, a satisfied invariant passes the candidate through; a violated one is a hard **REFUSE** (never negotiated, never granted post-hoc). Source:

`eres_trilogy_FINAL.py`, class `LockBox`.

Gate C — Pre-Qualification Inversion (*reference; already in force*)

Invariant	Cells are pre-qualified <i>before</i> access, not after.
Admit	$\text{requested_cells} \subseteq \text{accessible_cells}$
Refuse	$\text{any } (\text{requested_cell} \notin \text{accessible_cells}) \rightarrow \text{LockViolation("C")}$
Prevents	post-hoc access grants; negotiated escalation
Status	Demonstrated (trilogy test [14]; the gate that fires)

This is the pattern A, B, and D are formalized to match.

Gate A — Underwriting Direction (*formalize*)

Invariant	Infomediator underwrites; Infomediary is underwritten. Risk flows down; authority flows up.
Admit	Infomediator and Infomediary are both named AND distinct parties
Refuse	$\text{either party unnamed} \rightarrow \text{LockViolation("A")}; \text{OR } (\text{Infomediator} == \text{Infomediary}) \text{ (self-underwriting)} \rightarrow \text{LockViolation("A")}$
Prevents	self-underwriting; unattributed underwriting; inversion of the risk/authority direction

Gate B — Five-Layer Non-Extensibility (*formalize*)

Invariant	Data-Integrity has exactly five chiasmic-verification layers, in canonical order.
Canonical layers	$(\text{SEPLTA, H2C2H-C2H2C-binary, App-Parent-symmetry, binary-palindrome, IPIDITIS-IDIPITIS-transposition})$
Admit	$\text{layers} == \text{canonical five-tuple} \text{ (count = 5 AND ordering canonical)}$
Refuse	$\text{len(layers)} \neq 5 \rightarrow \text{LockViolation("B")}; \text{OR ordering} \neq \text{canonical} \rightarrow \text{LockViolation("B")}$
Prevents	adding/removing layers; re-opening the canonical Data-Integrity seal

Gate D — IOF Trinity Three-Register Grounding (*formalize*)

Invariant	The Trilogy is exactly three volumes grounded in three registers. No collapse, no addition.
Canonical volumes	$(\text{"OG"}, \text{"SC"}, \text{"DI"}) = \text{Biological (One-Good)} \cdot \text{Socio (Security-Clearance)} \cdot \text{Technical (Data-Integrity)}$
Admit	$\text{volumes} == (\text{"OG"}, \text{"SC"}, \text{"DI"})$
Refuse	any other tuple $\rightarrow \text{LockViolation("D")}$
Prevents	collapsing the three registers into fewer; adding a fourth register

(Gate E — SCALULAR Class Grammar, four pillars
HEALTH/LAW/PROTECTION/SKILLS_TRADE — exists in source but is out of scope for this addendum.)

2.1 Gate family vs. D(t)

The gates (Part 2) are **boundaries** — discrete, conjunctive, REFUSE-on-violation. D(t) (Part 1) is a **rating** — continuous, describing how far from coherence a state has drifted. Per the split principle, a rising D(t) never *becomes* a gate refusal on its own; it can only inform the tuning of gate parameters. The two are recorded separately.

Open items carried to v2

- 1. Per-channel coherence metric sub-specs (D(t) §1.1).
- 2. Integer weight vector w_k and W_sum.
- 3. θ_D, D_min, and the early-warning estimator window.
- 4. ARI / ERI / RCI canonical expansions (still open from BERA v1 §5).
- 5. Confirm Gate E inclusion in the canonical gate family or keep it trilogy-local.

Three Governing Principles (operationally binding): Don't hurt yourself. Don't hurt others. Build for generations to come.